

# Ayman Saissi

aymanessaissi@gmail.com | +44 7454789271 | [www.linkedin.com/in/aymane-saissi-658b7427b](https://www.linkedin.com/in/aymane-saissi-658b7427b)

**Portfolio:** <https://aymane-18.github.io/aymanessaissi.github.io/>

## EDUCATION

**The Cooper Union for the Advancement of Science and Art**, New York, NY

Graduated 2025

Bachelor of Engineering, *Mechanical Engineering*, **GPA 3.99/4.0** — **Dean's List all semesters**

- Awards: Innovator Merit Scholarship 2021-25, Half Tuition Scholarship 2021-24, Full Tuition Scholarship 2024-25, Jacqueline Bernstein and Marvis Scholarship 2021-25
- Tau Beta Pi Engineering Honor Society Member

**University of Cambridge**, Cambridge, UK

2025 - Present

PhD of Engineering, *Mechanical Engineering*

## RESEARCH EXPERIENCE

**Research Assistantship in Fluid Dynamics**, Cooper Union, New York, NY

Fall 2022 - Spring 2025

*Research Assistant for Flow Optimization*

- Designed specialized **nozzles and pipe systems** using **SolidWorks and Autodesk** for optimized flow performance
- Applied **Python-based data visualization** to assess and refine nozzle designs, minimizing turbulence and enhancing flow transitions
- Conducted **CFD simulations in Ansys** to analyze subsonic and hypersonic flow, incorporating variations in angle of attack and fluid properties
- Validated theoretical models through **high-speed imaging and pressure sensors**, capturing real-time flow dynamics and shock wave behavior
- Developed expertise in **turbulence modeling, boundary layer analysis, and compressible flow simulations**, optimizing aerodynamic performance

**Summer Exchange Program in Mechatronics**, Technische Universität Dresden, Dresden, Germany

Summer 2023

*Mechatronics Engineering Independent Researcher*

- **Designed** and executed **experiments** to analyze electric motor performance and durability
- Developed a predictive **machine learning framework** in Python, leveraging vibration data for early fault detection
- Engineered **SolidWorks**-based motor connections for optimized weight distribution and experimental accuracy
- Implemented **Node.js, Node-Red, and Javascript** for streamlined data processing and **UI (User Interface) development**
- Contributed to a **predictive model** for motor failure classification, improving reliability

## PROJECTS

**Adaptable Acoustic Panel for multi-Use Spaces**, Cooper Union, New York, NY

Fall 2024 - Spring 2025

- Led the design, fabrication, and testing of a mechanically switchable acoustic panel combining absorption and tunable diffusion
- Applied **ASME** acoustic testing **protocols (ISO 17497-2)** in controlled environments using RT60 and SPL measurements
- Gained 1 year of hands-on experience in a hardware engineering lab: **prototyping, CAD design** (SolidWorks), **manual fabrication**, and iterative testing
- Validated theoretical acoustic models (Sabine, Eyring, Fitzroy) with **Python** simulations and high-fidelity experimental data
- Created a customer-facing website using **HTML, JavaScript, and CSS**; version-controlled using Git and published via **GitHub Pages**
- Developed a low-cost, scalable product tested in classroom and anechoic chamber environments

**Drone Monitoring and Control**, Cooper Union, New York, NY

Fall 2021 - Spring 2022

- Used **Robot Operating System (ROS), Linux, and Python** for synchronized operation of multiple drones for collaborative tasks
- Utilized **Vicon cameras** and open-source drone technology to **monitor** and analyze the **motion** of the drone swarm
- Optimized swarm behavior and enhance efficiency in task execution in collaborated with multidisciplinary teams
- Coded in **Python** and **ROS** to implement navigation and obstacle avoidance, using **GitHub** for version control, collaboration, and issue tracking
- **Repaired and maintained drones**, including **hardware** replacement and **software updates**

## WORK EXPERIENCE

**Con Edison**, New York, New York

Summer 2024 - Summer 2025

*Co-op Engineer*

- Designed and implemented **Python** algorithms to **automate** the **extraction** of Negative Revenue Adjustments (NRAs) from multiples files
- Developed **predictive models** using linear regression to **forecast** key **business metrics** and drive data-driven decision-making
- Built an **interactive Power BI dashboard** to visualize key audit and NRA-related data, **tracking team progress** and streamlining reporting
- Gained **insights** into the **energy production and distribution system** in the **New York area**, enhancing understanding of utility operations and regulatory frameworks

## SKILLS

**Computer Programs:** Python, C++, Node.js, Node-Red, Linux, Microsoft, Mac, Final Cut Pro, Microsoft Office, SolidWorks, Onshape, Siemens NX, Matlab, Microchip Studio, MPLAB X IDE, Ansys, Github, Docker, AWS, Javascript, CSS.

**Fabrication:** 3D Printing, basic wood construction, Soldering.

**Languages:** Fluent in French, Arabic, and English. Beginner in Spanish.

## RESEARCH PUBLICATIONS AND CONFERENCE PRESENTATIONS

- A. Saissi, G. Sidebotham, K. Wright, I. Feier, "Workshops for Active Learning and the Draining Tank: A Low-Cost Thermal-Fluid Experiment", **ASME 2025 International Mechanical Engineering Congress and Exposition (IMECE2025)**, (Submitted for Technical Paper Publication, March 2025).

- A. Saissi, "Condition Monitoring of Induction Motor Using Vibration Signals and Machine Learning Classification", **2024 National Conference on Undergraduate Research (NCUR)**, Long Beach, CA, April 8–10, 2024.